Geostatistical Analysis of Post-Seismic Displacement Along the Ramapo Fault Using Continuous GPS Displacement Rates in New Jersey

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Abstract:

The Ramapo fault, extending through northern New Jersey and into adjacent states, represents a critical seismic boundary whose long-term stability has significant implications for the region's infrastructure and safety. On April 5, 2024, a major earthquake with a magnitude of 4.8 shook the area, offering a rare opportunity to evaluate the fault's present-day motion and its influence on surrounding landscapes. This study examines how the earthquake may have altered displacement patterns across the fault zone by analyzing continuous GPS records from stations positioned both east and west of the Ramapo fault. By incorporating each station's full operational timeline rather than restricting the analysis to a fixed observation window, we captured long-term deformation trends alongside short-term post-seismic adjustments. Using a GIS-interpolated map of average displacement rates, we identified localized zones of vertical and horizontal motion that correspond to distinct structural and geological settings along the fault. Preliminary results indicate differential movement across the fault, suggesting uneven strain release and possible shifts in subsurface stress distribution following the April 2024 event. These patterns have important implications for infrastructure resilience, as concentrated displacement zones may elevate risks of subsidence, slope instability, and structural damage. Findings reinforce the value of integrating geodetic monitoring into regional planning efforts to inform seismic hazard mitigation and guide development in this densely populated region.

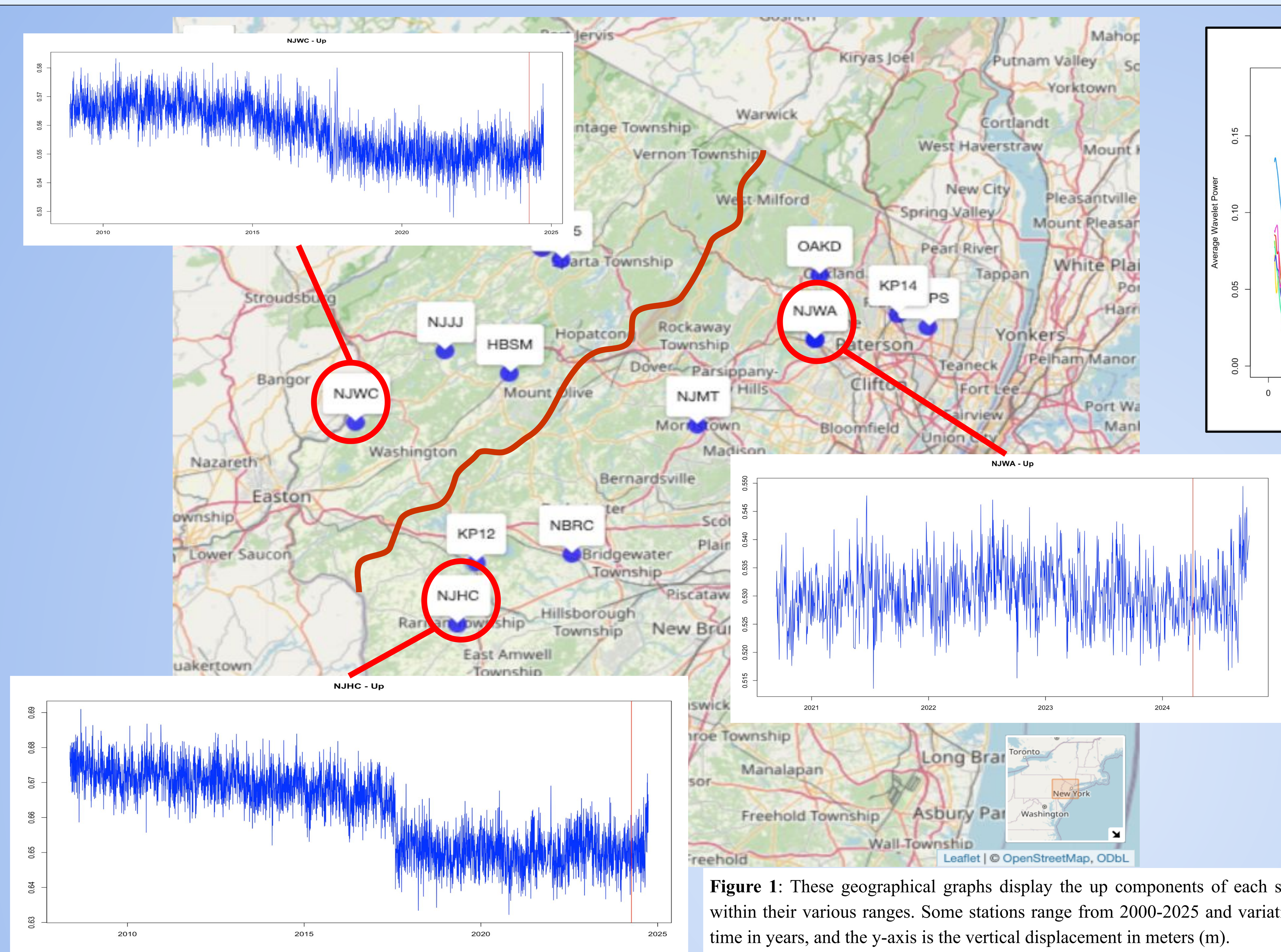

Figure 1: These geographical graphs display the up components of each station and their vertical movements within their various ranges. Some stations range from 2000-2025 and variations within the 2010s. .The x-axis is time in years, and the y-axis is the vertical displacement in meters (m).

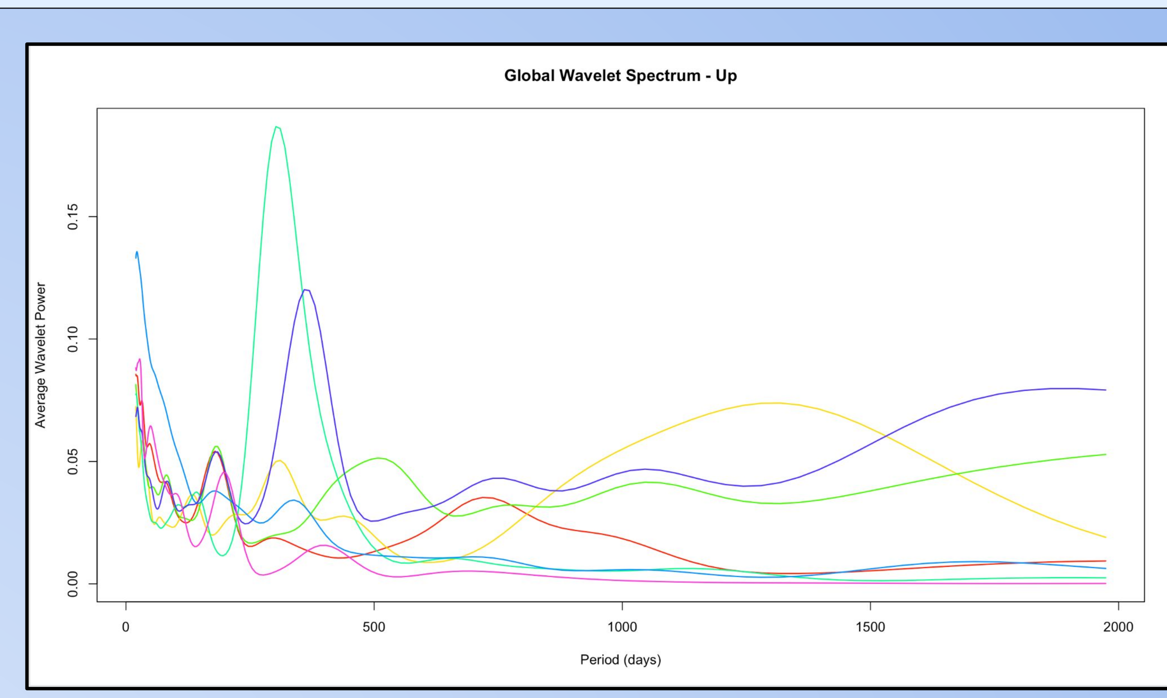

Figure 4: Global Wavelet Spectrum of the vertical (Up) component for selected GPS stations near the Ramapo fault, generated using the WaveletComp package. Showcases distribution of wavelet power across different periodicities, highlighting dominant oscillation frequencies in vertical displacement

Method/Materials

- GPS displacement data from 14 NJ stations were retrieved using R from the Nevada Geodetic Laboratory
- Data processing and cleaning used R packages (data.table, dplyr, lubridate, WaveletComp, gstat, leaflet)
- Custom function (GPS_dwd) structured data into northward, eastward, and upward displacement components.
- Missing values were interpolated, and stations with data outside scope were excluded (e.g. NJJJ, HBSM)
- Noise reduction applied using Kolmogorov-Zurbenko (KZ) filtering (365-day window, 3 replications)
- Wavelet Spectrum (WaveletComp) detected periodic signals in Up displacement components
- Linear regression (lm function) calculated vertical displacement rates from smoothed data
- Geostatistical interpolation (IDW method) mapped subsidence patterns using the gstat and sp packages.
- GIS visualization (leaflet package) created an interactive map showing GPS station locations, the interpolated vertical-displacement rate field, and the Ramapo fault trace, with pop-ups for station metadata

Background Information:

- The Ramapo Fault is a major geological boundary running through northern New Jersey into New York and Pennsylvania.
- It has long been recognized as a potential seismic hazard, though historically associated with low to moderate seismic activity.
- On April 5, 2024, a magnitude 4.8 earthquake struck the region - one of the strongest in recent local history.
- The Ramapo fault cuts across regions with dense infrastructure and population, increasing the importance of monitoring its behavior.
- Earthquakes can produce subsidence (land sinking) or uplift (land rising), reshaping landscapes and stressing infrastructure.
- Even moderate seismic events can redistribute subsurface stress, creating uneven ground motion across fault zones.
- Continuous GPS monitoring provides precise measurements of land deformation, capturing both long-term tectonic drift and short-term seismic shifts.
- GIS-interpolated maps highlight localized subsidence patterns, linking natural and human processes to elevated risks of ground instability and infrastructure vulnerability.

Results:

- **Geostatistical Analysis**
  - Interpolated vertical rates (m/yr) show a pronounced cross-fault gradient: a compact uplift cell (~+2 mm/yr) west of the Ramapo trace near Mt. Olive surrounded by broad subsidence (~-2 mm/yr) to the east/southwest (Fig. 1)
  - Gradient coincides with the fault corridor, indicating differential vertical motion across blocks rather than uniform regional drift
- **Subsidence/Uplift Trends**
  - Localized deformation: spatially confined hotspots/lows imply site-dependent response (likely mix of post-seismic stress changes, basin compaction, and loading effects)
  - Network response at event time: No coherent, network-wide permanent vertical step at April 5, 2024; any offsets are small and station-specific (Fig. 2)
- **Stations Observations**
  - NJWA (East Ramapo): No abrupt coseismic step on April 5; shows small, smooth uplit through late-2024
  - DES5 (NW of fault): Stable across event, prior spikes are earlier
  - NJSC (West of fault): No permanent offset, fluctuations after April 2024 stay within typical noise band
- **Wavelet Spectrum Analysis (Up)**
  - Dominant annual (~365-day) peak at most stations, consistent with seasonal loading
  - Secondary power at ~180 day and multi-year (≥1000 days) at select stations indicates slow background subsidence/uplift superimposed on seasonality (Fig. 3)
  - High-frequency power (<~50 days) is low after cleaning -> short-period noise largely suppressed

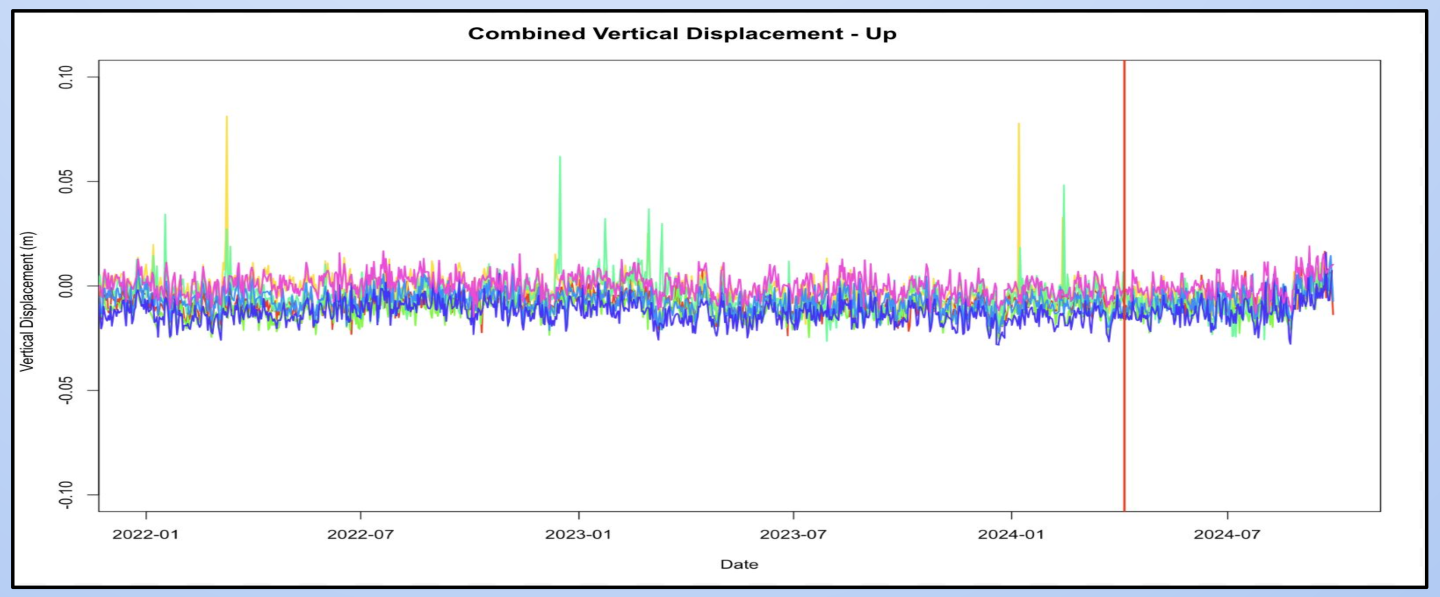

Figure 2: Combined vertical displacement (Up component) across all GPS stations near the Ramapo fault from 2022-2024. The solid red line marks the April 5, 2024 earthquake. No abrupt coseismic effect is observed, though small fluctuations suggest localized, transient adjustments. Created in RStudio.

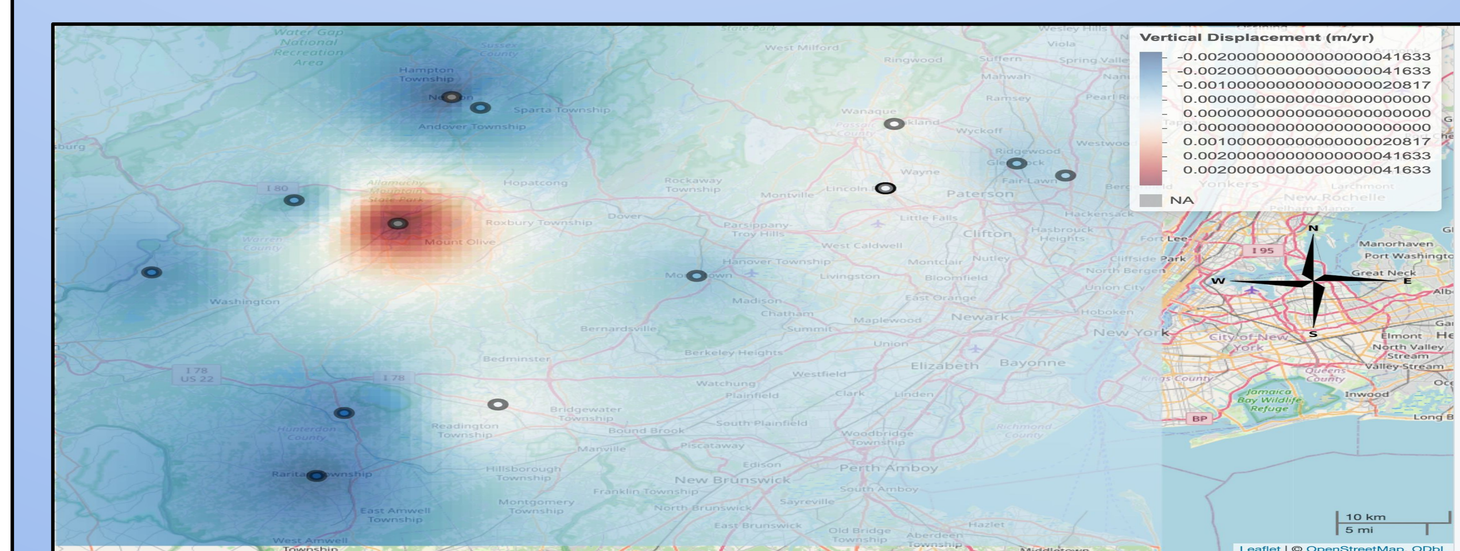

Figure 3: This GIS map displays the vertical displacement rates of each station and reveals potential areas of subsidence or uplift. The light red and white colored areas reveal areas of lower subsidence, the dark red colored areas reveal areas of uplift, and the blue colored areas reveal areas of high subsidence. Interactive map created in RStudio.

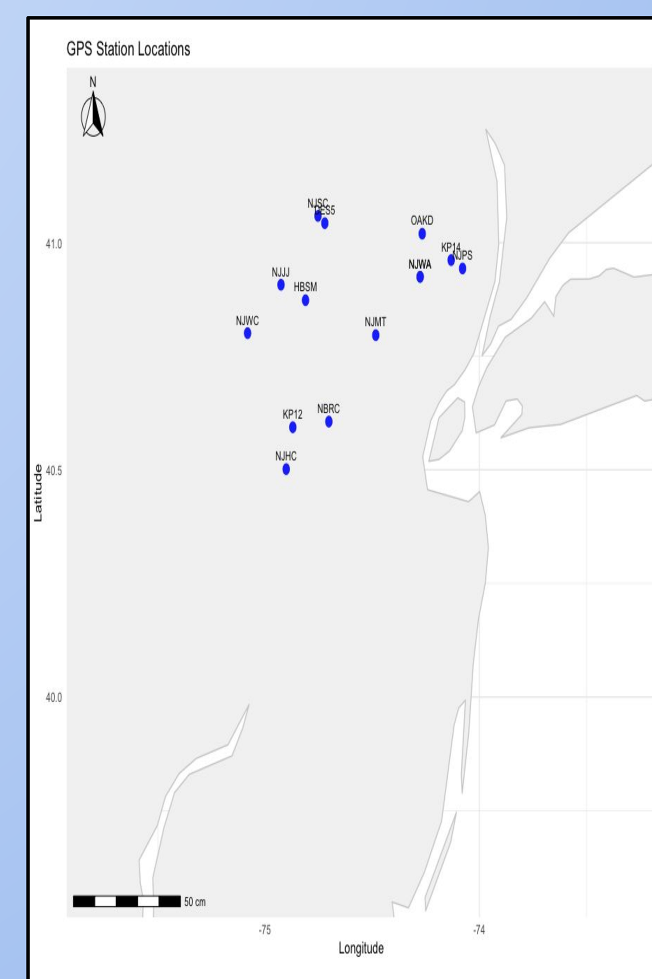

Figure 5: Locations of the 14 continuous GPS stations across northern New Jersey used to analyze vertical and horizontal displacement near the Ramapo fault.

Discussion/Conclusion

- Localized displacement patterns across the Ramapo fault show small, heterogeneous changes, with no large, permanent coseismic offsets observed after the April 5, 2024 earthquake
- Combined displacement time series indicates that observed fluctuations near the event are within seasonal or transient variability, not a clear step-function shift
- Station-specific response (e.g. NJWA, DES5, NJSC) suggest spatially variable post-seismic behavior, possibly tied to fault geometry and local geology
- Wavelet analysis highlights periodic oscillations consistent with annual and sub-annual cycles, confirming that seasonal signals dominate over short-term earthquake effects
- GIS-interpolated maps reveal zones of subsidence and uplift, indicating differential strain release across the fault rather than uniform motion.
- Findings suggest the April 2024 earthquake redistributed stress locally, but did not generate a corridor-wide vertical displacement event.
- Results emphasize the importance of continuous geodetic monitoring for identifying subtle, site-specific responses to seismic events and for guiding infrastructure resilience planning in northern NJ.

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